

EXPERIMENT 21

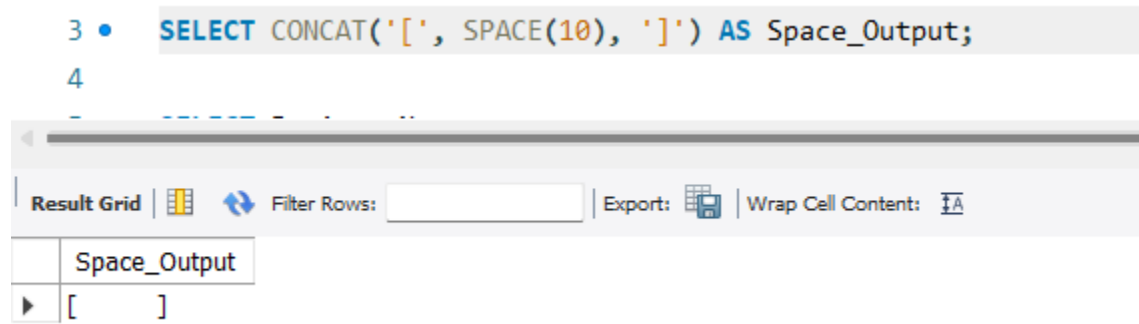
MySQL String Functions – SPACE, SUBSTR, UPPER, LOWER, TRIM, LENGTH

Aim: To implement SPACE(), SUBSTR(), UPPER(), LOWER(), TRIM(), and LENGTH() string functions using MySQL.

SPACE()

MySQL SPACE() returns the string containing a number of spaces as specified in the argument. Syntax : SPACE(N)

```
3 • SELECT CONCAT('[', SPACE(10), ']') AS Space_Output;
4
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Space_Output
[]

SUBSTRING() / SUBSTR()

MySQL SUBSTRING() returns a specified number of characters from a particular position of a given string. Syntax : SUBSTRING(str,pos,len)

MySQL SUBSTR() returns the specified number of characters from a particular position of a given string. SUBSTR() is a synonym for SUBSTRING().

Syntax : SUBSTR(str,pos,len)

```

5 • SELECT Patient_Name,
6       SUBSTR(Patient_Name, 2, 3) AS Substr_Output
7 FROM Patient;

```

Patient_Name	Substr_Output
Arun Kumar	run
Priya	riy
Rahul	ahu
Sneha	neh

UPPER()

MySQL UPPER() converts all the characters in a string to uppercase characters. Syntax : UPPER(str)

```

10 • SELECT Patient_Name,
11        UPPER(Patient_Name) AS Upper_Output
12 FROM Patient;

```

Patient_Name	Upper_Output
Arun Kumar	ARUN KUMAR
Priya	PRIYA
Rahul	RAHUL
Sneha	SNEHA

LOWER() /LCASE()

MySQL **LCASE()** converts the characters of a string to lower case characters.

Syntax : LCASE(str)

MySQL **LOWER()** **converts all** the characters in a string to lowercase characters.

Syntax: **LOWER** (str);

```

16 • SELECT Patient_Name,
17         LOWER(Patient_Name) AS Lower_Output
18 FROM Patient;

```

Patient_Name	Lower_Output
Arun Kumar	arun kumar
Priya	priya
Rahul	rahul
Snaha	snaha

TRIM()

MySQL TRIM() function returns a string after removing all prefixes or suffixes from the given string. Syntax : TRIM([{BOTH | LEADING | TRAILING} [remstr] FROM] str)

```

21 • SELECT TRIM(' Patient Name ') AS Trim_Output;

```

Trim_Output
Patient Name

RTRIM()

MySQL RTRIM() removes the trailing spaces from a given string. Syntax : RTRIM(str)

```
25 • SELECT CONCAT('[', RTRIM('Patient Name '), ']) AS RTrim_Output;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

RTrim_Output
[Patient Name]

LTRIM(str)

MySQL LTRIM() removes the leading space characters of a string passed as argument. Syntax : LTRIM(str)

```
28 • SELECT CONCAT('[', LTRIM(' Patient Name'), ']) AS LTrim_Output;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

LTrim_Output
[Patient Name]

LENGTH()

MySQL LENGTH() returns the length of a given string. Syntax :
LENGTH(str)

```
29 • SELECT Patient_Name,
30        LENGTH(Patient_Name) AS Length_Output
31 FROM Patient;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Patient_Name	Length_Output
▶	Arun Kumar	10
	Priya	5
	Rahul	5
	Sneha	5

BIT_LENGTH()

Returns the
length of the
string str in bits.

Syntax :

BIT_LENGTH(
str)

```
34 • SELECT Patient_Name,  
35       BIT_LENGTH(Patient_Name) AS Bit_Length  
36 FROM Patient;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content		
	Patient_Name	Bit_Length
▶	Arun Kumar	80
	Priya	40
	Rahul	40
	Sneha	40

CHAR_LENGTH()

Returns the length of the string str, measured in characters. A multi-byte character counts as a single character. This means that for a string containing five 2-byte characters, LENGTH() returns 10, whereas CHAR_LENGTH() returns 5.

```
39 • SELECT Patient_Name,  
40       CHAR_LENGTH(Patient_Name) AS Character_Length  
41 FROM Patient;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 		
	Patient_Name	Character_Length
▶	Arun Kumar	10
	Priya	5
	Rahul	5
	Sneha	5

CONCAT()

Returns the string that results from concatenating one or more arguments. If all arguments are nonbinary strings, the result is a nonbinary string. If the arguments include any binary strings, the result is a binary string. A numeric argument is converted to its equivalent nonbinary string form.

Syntax : CONCAT(str1,str2,...)

```
44 • SELECT Patient_Name,  
45         Disease,  
46         CONCAT(Patient_Name, ' - ', Disease) AS Combined_Output  
47 FROM Patient;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Patient_Name	Disease	Combined_Output	
▶ Arun Kumar	Fever	Arun Kumar - Fever	
Priya	Diabetes	Priya - Diabetes	
Rahul	Asthma	Rahul - Asthma	
Sneha	Migraine	Sneha - Migraine	

INSTR()

MySQL INSTR() takes a string and a substring of it as arguments, and returns an integer which indicates the position of the first occurrence of the substring within the string

Syntax : INSTR(str,substr)

```
50 • SELECT Patient_Name,  
51         INSTR(Patient_Name, 'a') AS Position_Output  
52 FROM Patient;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Patient_Name	Position_Output		
▶ Arun Kumar	1		
Priya	5		
Rahul	2		
Sneha	5		

LOCATE()

MySQL LOCATE() returns the position of the first occurrence of a string within a string. Both of these strings are passed as arguments. An optional argument may be used to specify from which position of the string (i.e. string to be searched) searching will start. If this position is not mentioned, searching starts from the beginning.

Syntax:

LOCATE(substr,str,pos)

```
55 • SELECT Patient_Name,
56         LOCATE('a', Patient_Name) AS Locate_Output
57 FROM Patient;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Patient_Name	Locate_Output
▶	Arun Kumar	1
	Priya	5
	Rahul	2
	Sneha	5

MID()

MySQL MID() extracts a substring from a string. The actual string, position to start extraction and length of the extracted string - all are specified as arguments.

Syntax : MID(str,pos,len)

```
60 • SELECT Patient_Name,
61         MID(Patient_Name, 2, 3) AS Mid_Output
62 FROM Patient;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

	Patient_Name	Mid_Output
▶	Arun Kumar	run
	Priya	riy
	Rahul	ahu
	Sneha	neh

POSITION()

MySQL POSITION() returns the position of a substring within a string.. Syntax :

POSITION(substr IN str)

```
65 • SELECT Patient_Name,  
66         POSITION('a' IN Patient_Name) AS Position_Output  
67 FROM Patient;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Patient_Name	Position_Output
▶	Arun Kumar	1
	Priya	5
	Rahul	2
	Sneha	5

RESULT:

Thus the String Functions such as space, substr, upper, lower, trim, length are executed successfully.

